sequence, ignore the dog, cannon, tree, and cliff; focus solely on your windowpanes and the ball. As seen by you, relative to your windowpanes, the ball moves along the straight dashed line with constant velocity.

Thus, in the dog's reference frame the ball obeys Newton's laws; it moves along a parabola. In your small, freely falling reference frame it obeys the laws of gravity-free special relativity; it moves along a straight line with constant velocity. And what is true in this example must be true in general, Einstein realized in a great leap of insight:

In any small freely falling reference frame anywhere in our rea4 gravity-endowed Universe, the laws of physics must be the same as they are in an inertial reference frame in an idealized, gravity-free universe. Einstein called this the *principle of equivalence*, because it asserts that small, freely falling frames in the presence of gravity are equivalent to inertial frames in the absence of gravity.

This assertion, Einstein realized, had an enormously important consequence: It implied that, if we merely give the name "inertial reference frame" to every small, freely falling reference frame in our real, gravity-endowed Universe (for example, to a little laboratory that you carry as you fall over the cliff), then everything that special relativity says about inertial frames in an idealized universe without gravity will automatically also be true in our real Universe. Most important, the *principle of relativity* must be true: All small, inertial (freely falling) reference frames in our real, gravity-endowed Universe must be "created equal"; none can be preferred over any other in the eyes of the laws of physics. Or, stated more precisely (see Chapter 1):

Formulate any law of physics in terms of measurements made in one smal4 inertial (freely falling) reference frame. Then, when restated in terms of measurements in any other small inertial (freely falling) frame, that law of physics must take on precisely the same mathematical and logical form as in the original frame. And this must be true whether the (freely falling) inertial frame is in gravity-free intergalactic space, or is falling off a cliff on Earth, or is at the center of our galaxy, or is falling through the horizon of a black hole.